

SOH EE ZEN

Homepage: soheezen.github.io | LinkedIn: linkedin.com/in/soheezen

EDUCATION

National University of Singapore

Aug 2023 - Present

Bachelor of Science (Hons) (Current GPA: 4.85 / 5.0)

- Bachelor of Science in Mathematics
- Specialization in Pure Mathematics, Minor in Statistics
- Special Program in Mathematics (SPM)
- Dean's List - AY24/25 Sem 1, AY25/26 Sem 1, AY25/26 Sem 2

Yishun Innova Junior College

Feb 2019 - Oct 2020

Singapore-Cambridge GCE 'A' Levels (AAA/B, Merit)

- Subjects taken: H3 Maths, H2 Further Maths, H2 Maths, H2 Physics, H1 History

RESEARCH EXPERIENCE

Final Year Project, NUS

Aug 2026 - Present

- *Problems on Cluster Algebras*
Supervisor: Bao Huanchen

Research in Industrial Projects for Students (RIPS), IMS

May 2026 - Jul 2026

- Project Sponsor: NVIDIA
- Implemented an NVIDIA Warp-enabled FEM solver and a multilevel diffusion model for PDE inverse problems

Independent Study, NUS

Aug 2025 - Nov 2025

- *Abelian Categories and Derived Functors*
Supervisor: David Hansen
- Studied the theory of homological algebra over general abelian categories and applications to group theory

UG Research Opportunities Program in Science (UROPS), NUS

Aug 2024 - Apr 2025

- *Representation Theory of Symmetric Groups and the Combinatorial Invariance Conjecture* ([link](#))
Supervisor: Bao Huanchen
- Provided proofs for (1.6) and (1.7) in M. J. Dyer's *Hecke algebras and shellings of Bruhat intervals*, and compiled results related to the combinatorial invariance of the dihedral groups

TEACHING EXPERIENCE

Yio Chu Kang Secondary School, TeachSG Tutor

Mar 2025 - Sep 2025

- Tutor to a group of Secondary 4 students, giving guidance to their difficulties and providing extra materials to practice their knowledge, as well as planning other activities for their enrichment

Private Maths Tutor

Mar 2023 - Present

- Taught students following the GCSE O-Level and A-Level syllabus, helping them grasp complex mathematical concepts with improvements of at least 3 grades

RELEVANT COURSES TAKEN

- MA5204 Commutative and Homological Algebra, AY24/25 Sem 2
- MA5218 Representation Theory AY25/26 Sem 1
- MA4203 Galois Theory AY24/25 Sem 2

ADDITIONAL INFORMATION

- Programming Languages: Python, R, Latex